BrickBuilder: Phase-1 MVP

Product Requirements Document

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Table of Contents

1. Executive Summary & Product Vision

Overview

BrickBuilder is a web-based AI platform that transforms imagination into tangible reality. By combining cutting-edge generative AI, 3D visualization, and seamless e-commerce integration, BrickBuilder enables users to create custom LEGO-style builds from simple text or image prompts complete with step-by-step instructions, real part sourcing, and direct purchasing.

Demo Link:

www.image2brick.com

Target Audience

Primary Users:

> The target audience of the BrickBuilder MVP will be children aged 5-12 years of age. Once this market is captured in a growing manner, BrickBuilder will look to pivot into focusing on the MOC creators and LEGO enthusiasts.

Core Value Proposition

"From Imagination to Instructions in Seconds"

BrickBuilder eliminates the traditional friction between creative vision and physical creation. Users experience:

- **Instant Gratification:** AI generates buildable models in under 10 seconds
- **Zero Complexity:** No CAD skills, no part catalogs, no guesswork
- **Real-World Connection:** Actual LEGO parts with pricing and availability
- **Complete Solution:** From concept to doorstep in a single, magical flow

Emotional Experience

BrickBuilder must feel like magic meeting practicality. The moment a user types "spaceship" and watches a 3D model materialize before their eyes should evoke wonder, delight, and childlike excitement. Yet the transition from dream to reality from viewing the bill of materials, seeing real prices, completing checkout is grounded in a feeling of magic that turns into something tangible.

Phase-1 MVP Scope

This document defines a clickable web prototype (non-functional) that demonstrates the complete user journey without backend implementation. The goal is to create a visually stunning, interaction-complete prototype that:

- Validates the core UX flow and information architecture
- Serves as a high-fidelity blueprint for engineering
- Generates stakeholder excitement and investment confidence
- Tests user comprehension through usability sessions

What's In Scope:

- All primary screens and interactions (desktop-first)
- Visual design system and component library
- Simulated generation and transitions
- Static sample models, BOMs, and pricing

What's Out of Scope:

- Backend AI integration
- Real-time model generation
- Payment processing
- User accounts and authentication
- Mobile-responsive layouts (optional stretch goal)

2. Design Philosophy & User Experience Principles

The BrickBuilder Experience Philosophy

"Invisible Complexity, Visible Magic"

Every decision in BrickBuilder's design must serve a singular purpose: to make the complex feel simple, the technical feel intuitive, and the transactional feel delightful. We are not building software… we are building an experience that transforms how people bring ideas to life. We sell the experience of turning a dream into reality.

Core Design Principles

1. Immediacy Over Perfection

Users should see results instantly. A good-enough model in 5 seconds beats a perfect model in 60 seconds. Progressive refinement can follow, but the initial "wow" moment must be immediate.

Design Implication: Loading states are theatrical, not apologetic. Animations suggest active creation, not passive waiting.

2. Clarity Through Minimalism

Every screen should have one primary action and one clear next step. No overwhelming menus, no decision paralysis. The interface disappears, leaving only the creation.

Design Implication: White space is generous. Buttons are obvious. Information hierarchy is ruthless.

3. Playful Professionalism

BrickBuilder lives at the intersection of toy and tool. The interface should feel approachable and fun, but never frivolous or amateurish. Think: Apple's clarity meets LEGO's playfulness.

Design Implication: Colors are vibrant but sophisticated. Copy is friendly but confident. Interactions are delightful but purposeful.

4. Tangible Connection

Digital experiences often feel ephemeral. BrickBuilder must constantly remind users that their creation can become real. Part names, prices, weight, delivery estimates these details anchor the fantasy in reality.

Design Implication: BOMs feel like treasure maps. Prices build anticipation, not anxiety. Checkout confirms dreams.

5. Progressive Disclosure

Never show users everything at once. Each step reveals only what's needed now, creating a sense of natural flow and preventing overwhelm.

Design Implication: Advanced options are hidden until needed. The core path is a straight line. Complexity is available, never mandatory.

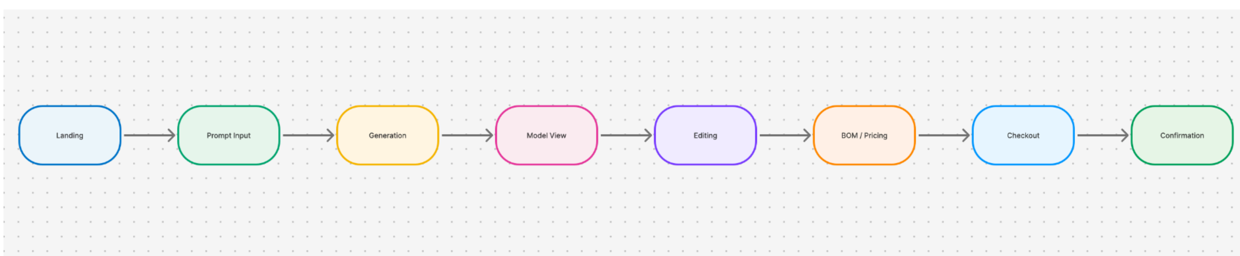
The Five-Second Test

Every screen must pass this test: *Can a first-time user understand what to do within five seconds?*

If yes, the design succeeds. If no, simplify.

Emotional Journey Map

Landing: Curiosity → "What is this?"
Prompt Input: Anticipation → "Let's see what happens"
Generation: Wonder → "It's actually creating something!"
Model View: Delight → "This is mine!"
Editing: Empowerment → "I can make it perfect"
BOM/Pricing: Excitement → "I can really build this"
Checkout: Commitment → "This is happening"
Confirmation: Satisfaction → "I made something real"



3. End-to-End UX Flow

Complete User Journey Overview

The BrickBuilder experience is designed as a linear narrative with optional side paths. Users move forward naturally, with the option to step back and refine at any point. The flow is optimized for first-time users who have never used the platform before.

Five Core Stages

Stage 1: Landing & Prompt Input

Goal: Inspire users and collect their creative input
Duration: 15-30 seconds
Key Actions:

1. User arrives at landing page
2. Reads headline and sees example builds
3. Types creative prompt into prominent input field (or uploads image)
4. Clicks "Generate" button
5. System validates input and initiates generation

Success Criteria: User understands the product value and feels confident submitting a prompt

Visual Cues (Landing Page)

- Hero Section: The central input bar is wide, centered, and visually dominant. Inside it, dynamic example prompts fade in and out ("Build a spaceship", "Make a castle", "Design your dream car"), subtly implying endless creative possibilities.

- Upload Icon: A small glowing image-upload icon sits at the right end of the input bar, hinting at multimodal creativity (text or image).

- Generate Button: The "Generate" button just below pulses softly with energy signaling action and immediacy without distraction.

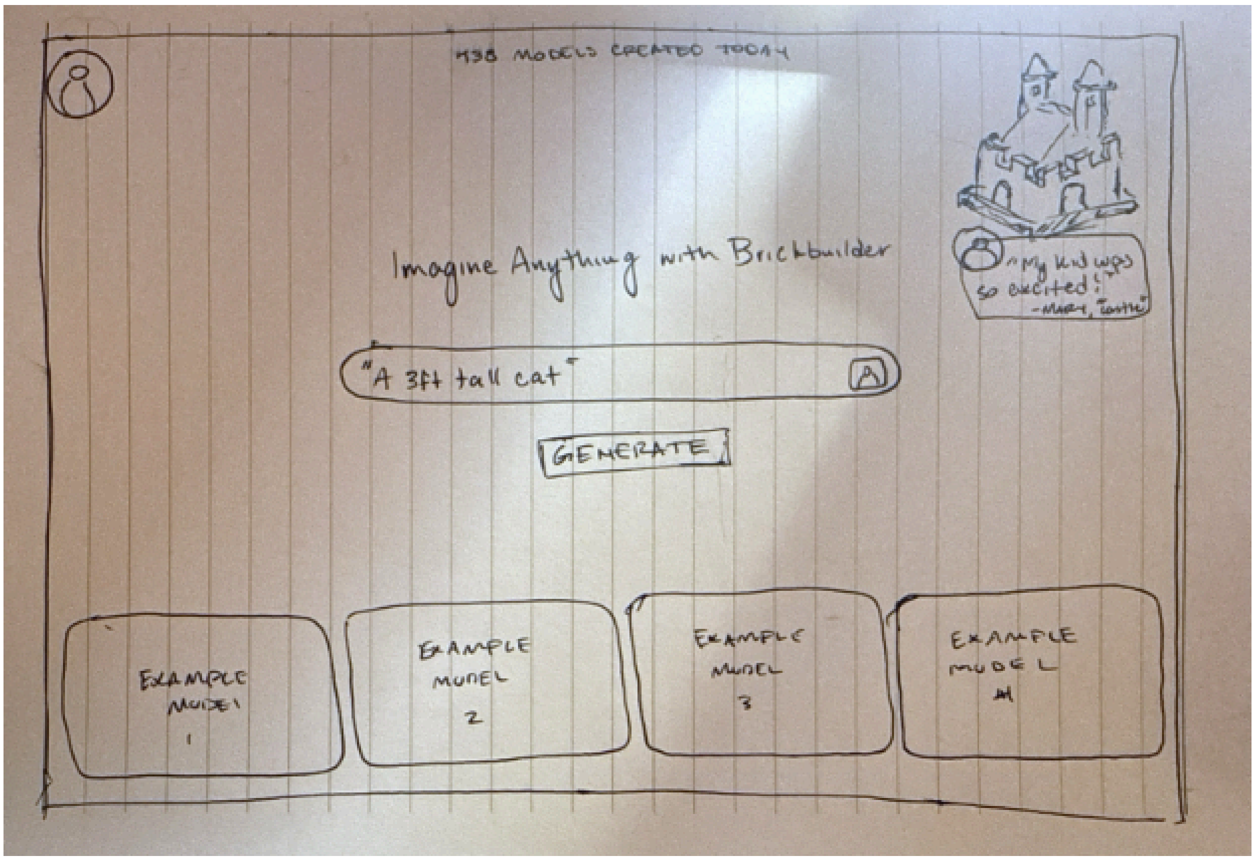
- Filters Row: Each filter (piece count, cost, size, steps, build time, theme) uses intuitive slider and dropdown styles, gently animating when hovered to suggest interactivity.

- 3D Showcase: In the top-right, a slowly rotating isometric model conveys "living creativity." A small testimonial bubble at its corner (avatar + quote) humanizes the experience and adds emotional credibility.

- Horizontal Gallery: The auto-scrolling gallery below the filters uses gentle motion, drawing the eye to example builds without clutter. Each card has clear visual hierarchy image first, then price, part count, and delivery time.

- Header Icons: The profile icon in the top-right subtly enlarges on hover, hinting at personalization.

- Footer: The bottom bar remains minimal Help, Contact, FAQ, and Terms links softly illuminate when hovered, maintaining the sleek futuristic aesthetic.



Stage 2: Generation & Loading

Goal: Build anticipation while AI creates the model
Duration: 8-12 seconds (simulated)
Key Actions:

1. Screen transitions to generation view
2. Loading animation plays (mechanical, building-themed)
3. Progressive messages update ("Analyzing prompt...", "Selecting bricks...", "Building model...")
4. Completion chime plays
5. 3D model fades in

Success Criteria: User remains engaged and excited during the wait

Visual Cues (Loading / Generation Page)

- Live Brick Construction Animation: A 3D LEGO model progressively assembles in real time, brick by brick, representing the exact design the user's AI model is generating. Each piece clicks satisfyingly into place with smooth easing and realistic physics, mirroring the building process.

- Progress Visualization: A horizontal progress bar or circular percentage counter accompanies the animation, visually syncing to the model's assembly. As the structure nears completion, the animation accelerates slightly to heighten anticipation.

- Atmospheric Motion Elements: Light ambient particles drift through the background—dust motes, sparks, or miniature LEGO studs giving the feeling of creation happening inside a digital workshop.

- Encouraging Microcopy: Brief, playful text rotates beneath the loading indicator to sustain engagement, e.g.:

    - "Gathering bricks from the digital warehouse…"

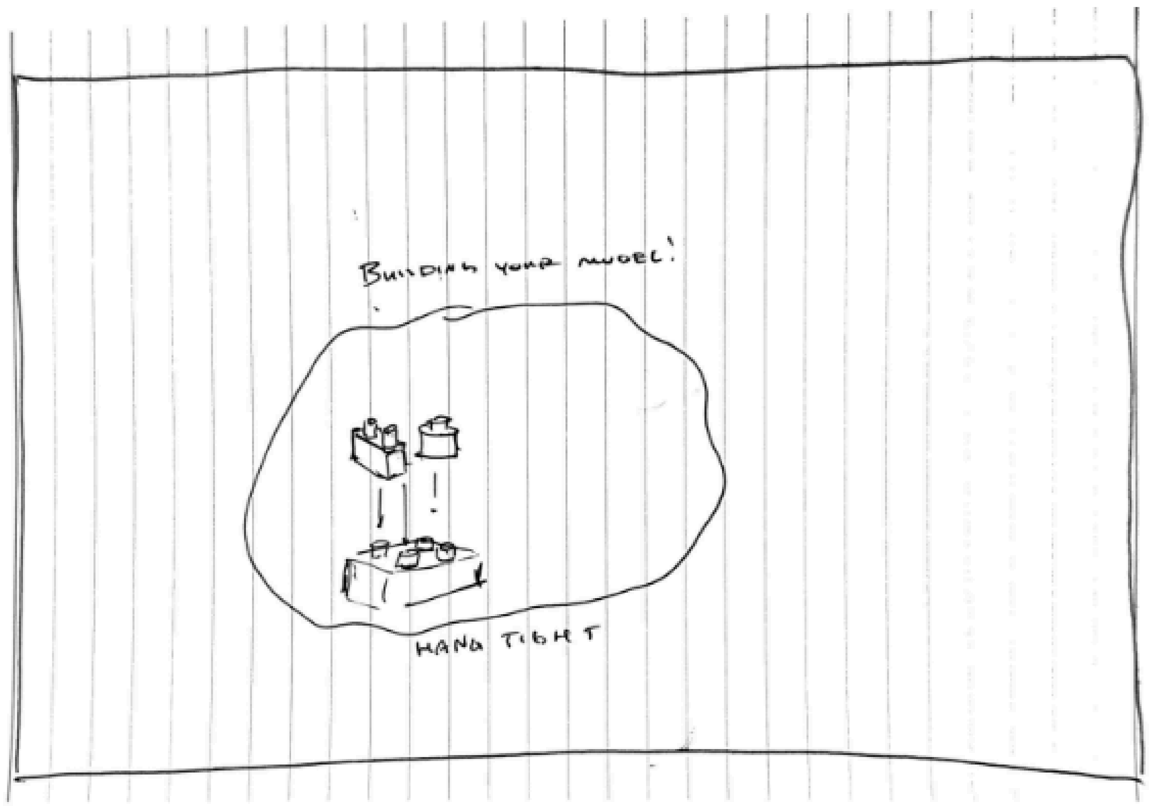
    - "Snapping your imagination together…"

    - "Aligning the last piece…"
      Each phrase is short, whimsical, and thematically tied to the joy of building.

- Subtle Sound Design (Optional): Soft clicking or clacking sounds accompany the virtual brick assembly. A faint synth swell or tonal hum reinforces a sense of creativity in motion without overwhelming the user.

- Visual Continuity: The overall palette remains consistent with the landing page cool futuristic tones, soft glows, and a focus on clean motion rather than clutter. When loading completes, the animation smoothly transitions into the Model Viewer by fading out the background and pulling the user seamlessly into the finished build

Technical Note: In prototype, this is a timed delay with pre-loaded model result



Stage 3: 3D Model Viewer & Interaction

Goal: Let users explore and fall in love with their creation
Duration: 1-3 minutes
Key Actions:

1. 3D model appears in interactive viewer
2. User rotates, zooms, and examines model
3. Model name and basic stats display
4. User clicks "Continue to Instructions" or "Edit Model"

Success Criteria: User feels ownership and pride in the generated model

Visual Cues (Model Generation / Preview Page)

- Hero Model Display: The 3D model sits prominently at the center of the screen, beautifully lit with soft ambient shadows and gentle bloom designed to make every brick gleam like a tangible object.

- Dynamic Animation: The model performs a slow, continuous rotation for the first few seconds to invite curiosity. Once the user interacts, rotation pauses automatically to give full control.

- Live Interaction Cues: Semi-transparent rotation arrows appear near the corners of the model's bounding area, subtly hinting that the user can rotate. Zoom icons and a small on-screen control wheel fade in on hover, making interactivity intuitive without clutter.

- Real-Time Model Stats Panel: A side or bottom information panel dynamically updates as the model is generated, showing key data:

    - Part Count: e.g., "342 pieces"

    - Estimated Build Time: e.g., "Approx. 45 minutes"

    - Estimated Shipping: e.g., "Parts ship in 3–5 days"

    - Instruction Steps: e.g., "92 steps total"

    - Estimated Total Cost: e.g., "$67.40 USD"
      Each metric uses small animated icons (brick, clock, truck, wrench, dollar symbol) for visual readability.

- Edit Mode Entry Point: A "Refine or Edit" button floats near the bottom corner, allowing the user to open an overlay where they can tweak the design—change colors, scale, or theme. Adjustments are visualized live, with the model regenerating in-place.

- Guided Attention: After 3–5 seconds of idle viewing, a prominent CTA ("Next: See Instructions & Pricing") animates into view, pulsing softly to draw the eye.

- Micro Motion Details: Ambient lighting gently shifts hues as the model sits idle (subtle color drift), creating a sense of life without distraction. Small sparkles or dust motes drift across the scene to simulate physical presence.

- Optional Social Element: In one corner, a small "Built by [username]" tag may appear with a miniature profile avatar if the design originated from a shared template reinforcing community and authenticity.

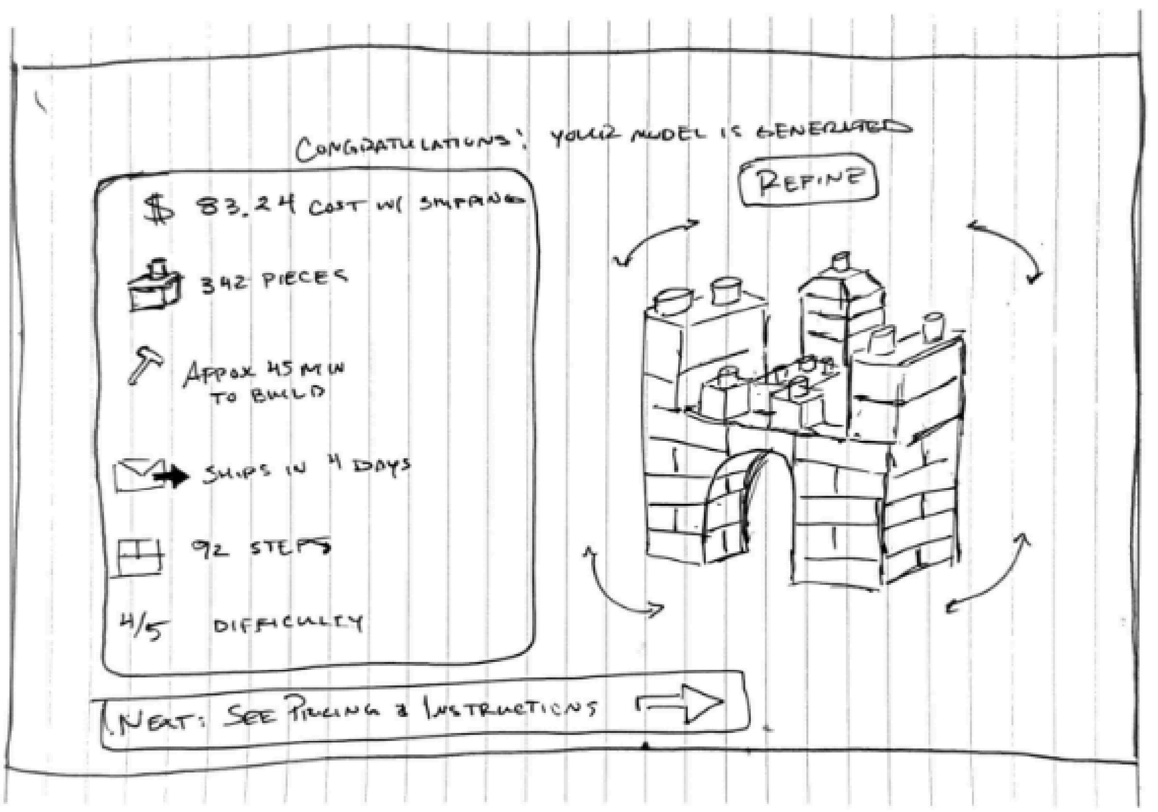
Interaction Patterns

- Click + Drag: Rotates the model around all axes.

- Scroll / Pinch: Smooth zoom in and out (limits enforced to avoid clipping).

- Double-Click: Resets the model to the default viewing position.

- Click Individual Bricks (Optional): Highlights that brick in soft glow; future functionality may reveal brick details or substitute options.



Stage 4: Edit & Refine Controls *(Optional Path)*

Goal: Give users basic customization without overwhelming them
Duration: 30 seconds - 2 minutes
Key Actions:

1. User accesses edit mode
2. Adjusts colors, scale, or complexity via simple sliders
3. Clicks "Regenerate" to apply changes
4. Views updated model
5. Either saves changes or reverts
6. Proceeds to BOM

Success Criteria: Adjustments feel powerful but simple; users don't get lost

Visual Cues (Edit & Refine Mode)

- Conversational Edit Interface: When users click *"Refine / Edit"*, the page subtly transitions into a focused dialog mode a text box appears beneath or beside the 3D model, inviting the user to type natural instructions such as:
  "Make it smaller," "Change the color scheme to red and black," or *"Add more windows."*
  This design keeps the process intuitive and free-flowing, more like speaking to a creative assistant than using design software.

- Smooth Context Transition: The model remains fully visible and gently shifts position to make space for the text input field, maintaining continuity and reinforcing that the edits affect this exact object in view.

- AI Response Feedback: After submission, the system shows a short message like *"Got it! Updating your build…"* followed by the model regenerating live in the same viewport, brick by brick, to visualize the change.

- Visual Continuity: A progress spinner or thin animated bar at the bottom of the text box indicates regeneration activity, ensuring users understand that the system is processing their request.

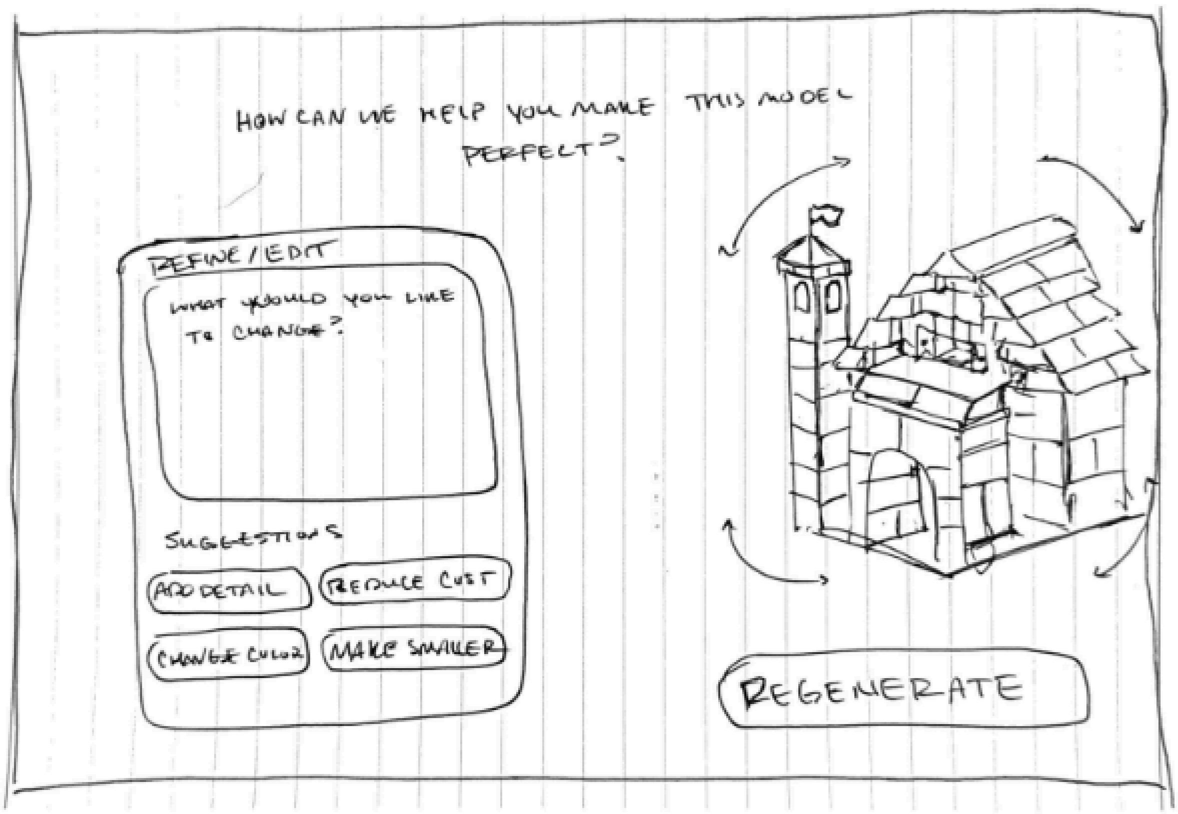
- Regeneration Confirmation: When complete, a brief glowing animation plays over the updated areas of the model, accompanied by a short confirmation message such as *"Model updated!"*

- Optional Quick Prompts: To guide users, several suggested edit commands appear as light grey chips above the text box (e.g., *"Add detail"*, *"Simplify"*, *"Change color"*, *"Make smaller"*).

- Exit Flow: Clicking *"Done"* or pressing *Escape* returns the user to the standard viewing mode. The edit box fades out, leaving the refreshed model and updated data panel visible.

- Emotional Tone: The goal is to make the user feel they're co-creating with an AI, not tweaking a tool simple, empowering, and magical.

Technical Note: For MVP, 2-3 preset variations simulate regeneration



Stage 5: BOM, Pricing & Checkout

Goal: Transform digital creation into tangible purchase
Duration: 1-2 minutes
Key Actions:

1. User clicks "Get Building Instructions"
2. System displays bill of materials with part images, names, quantities
3. Total price and availability status shown
4. User reviews parts list
5. Clicks "Purchase Parts" or "Download Instructions"
6. Completes checkout (email, shipping, payment)
7. Receives order confirmation

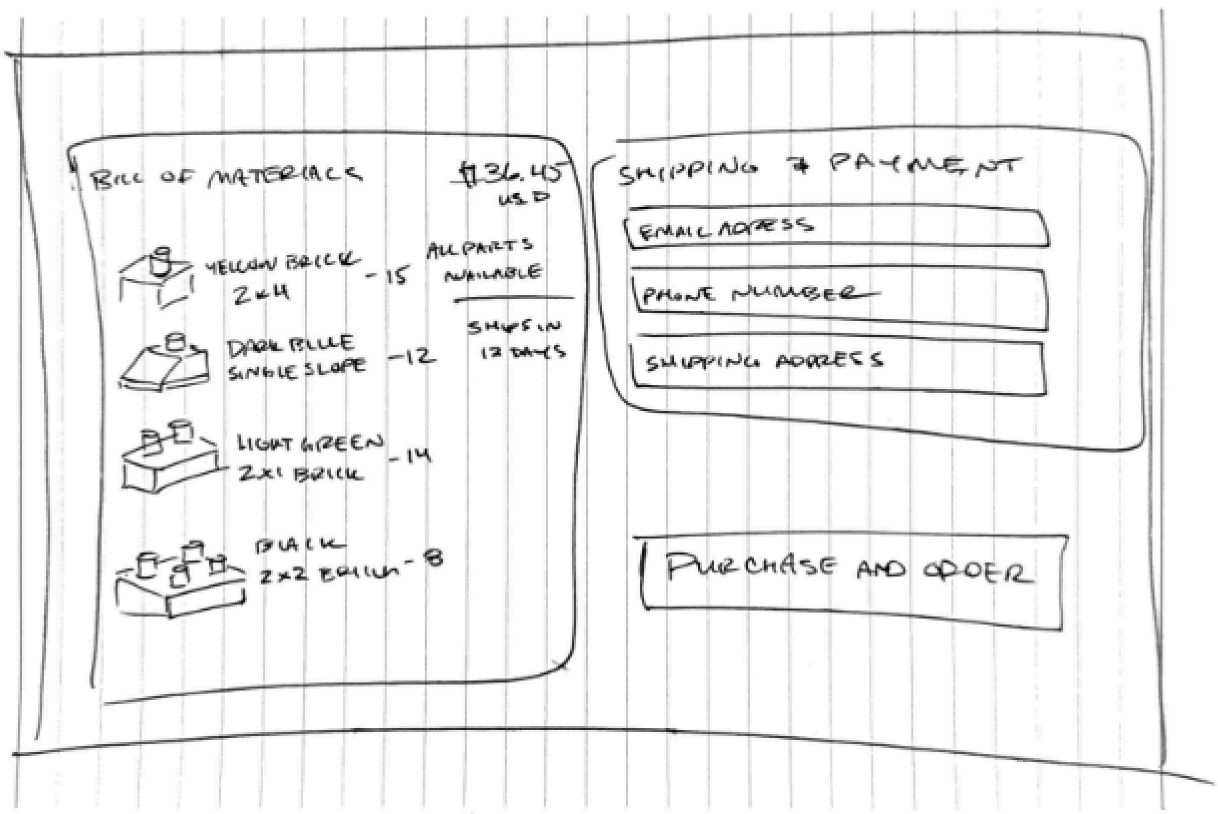
Success Criteria: Transition from free exploration to paid commitment feels natural and exciting, not transactional

Visual Cues:

- BOM is visually rich (part photos, colors)
- Pricing feels transparent and reasonable
- Checkout is streamlined (single page, minimal fields)
- Confirmation feels celebratory

Key Decision Points:

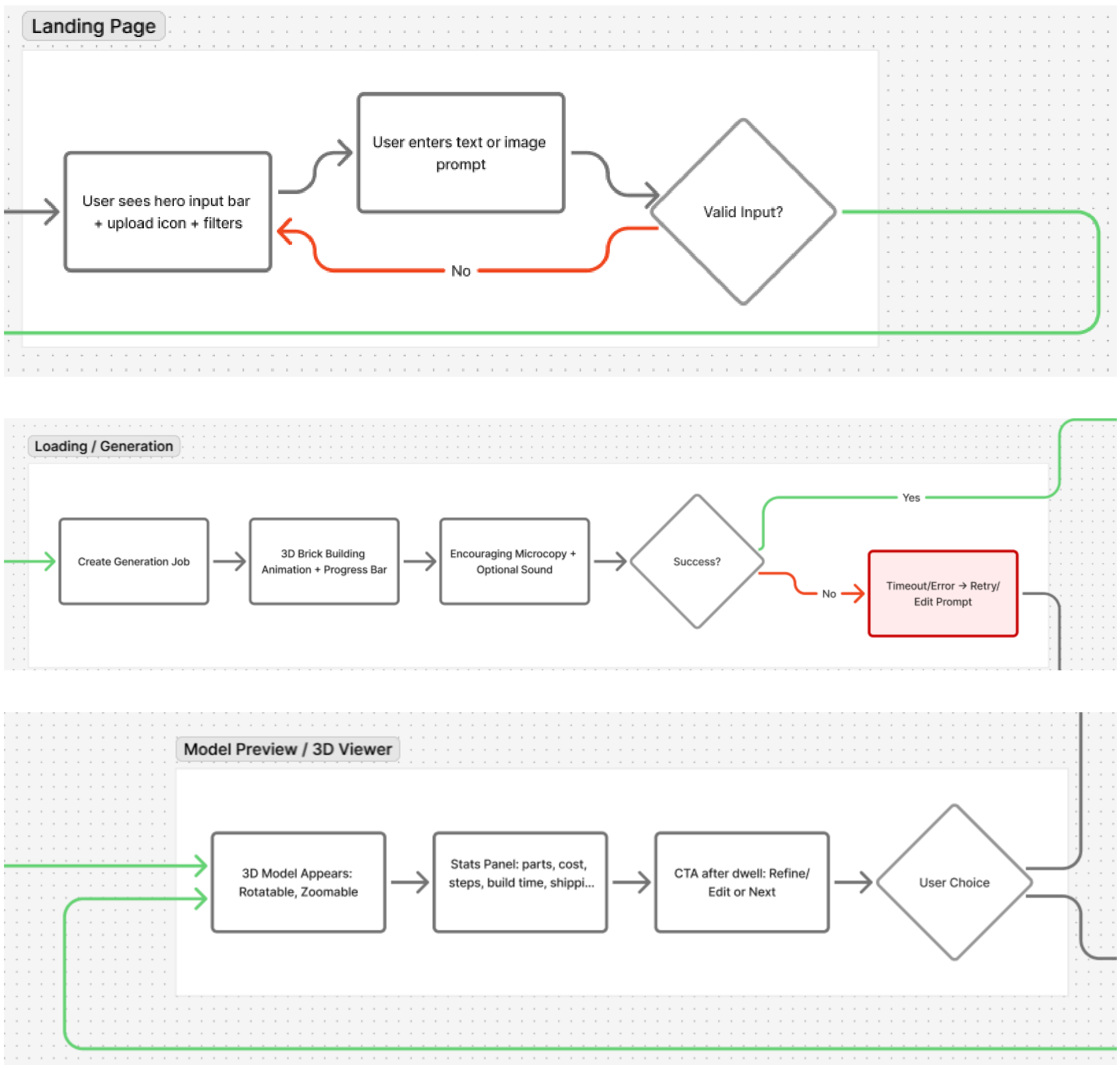
- Download free instructions (no purchase)
- Purchase full kit
- Purchase missing parts only (for users with existing LEGO)

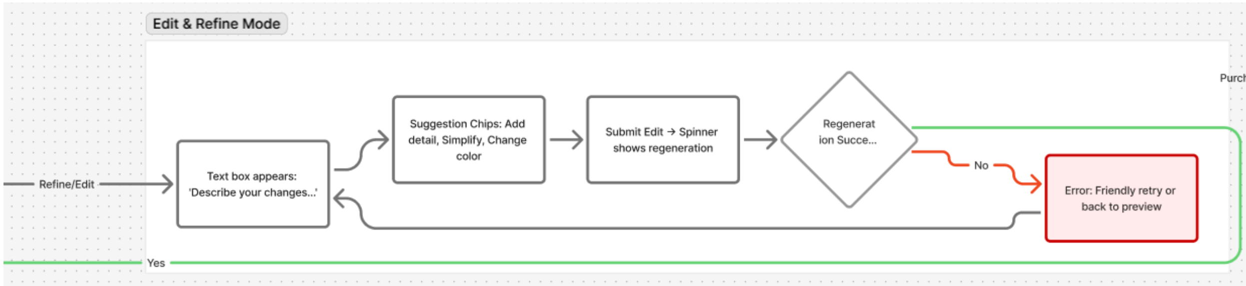


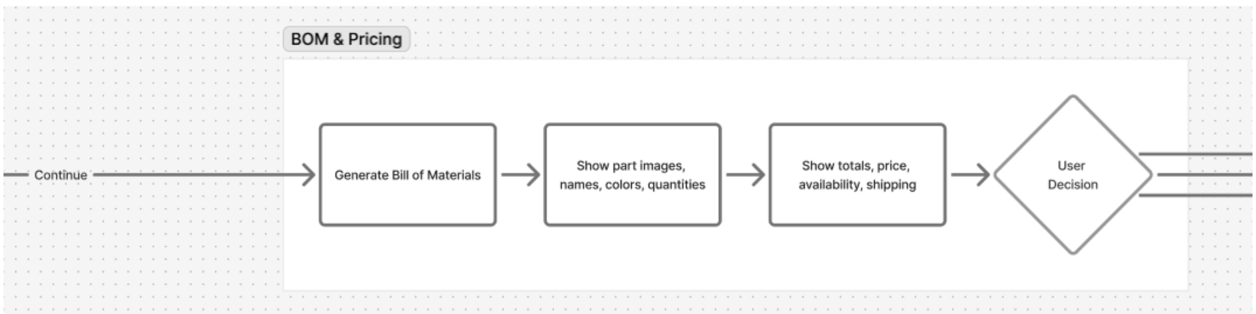
Complete Flow Diagram

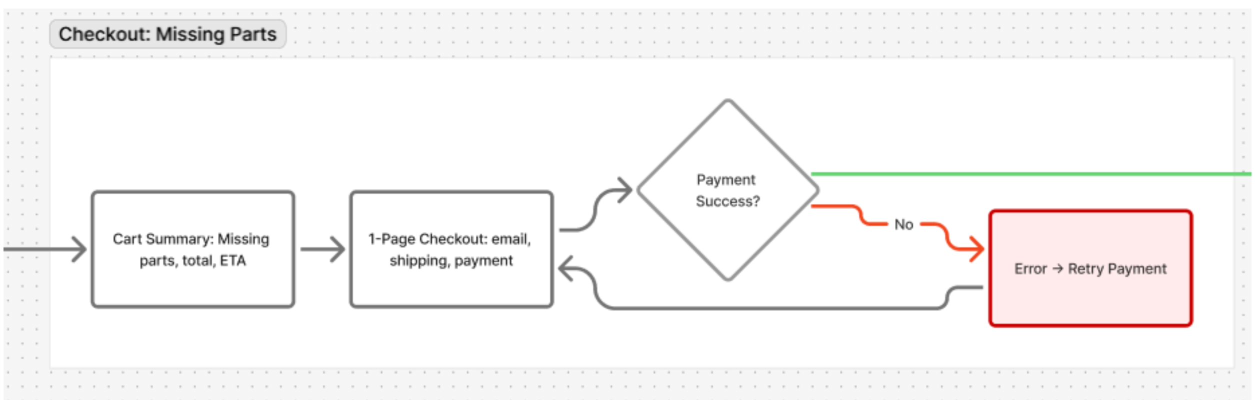
[Figma File]

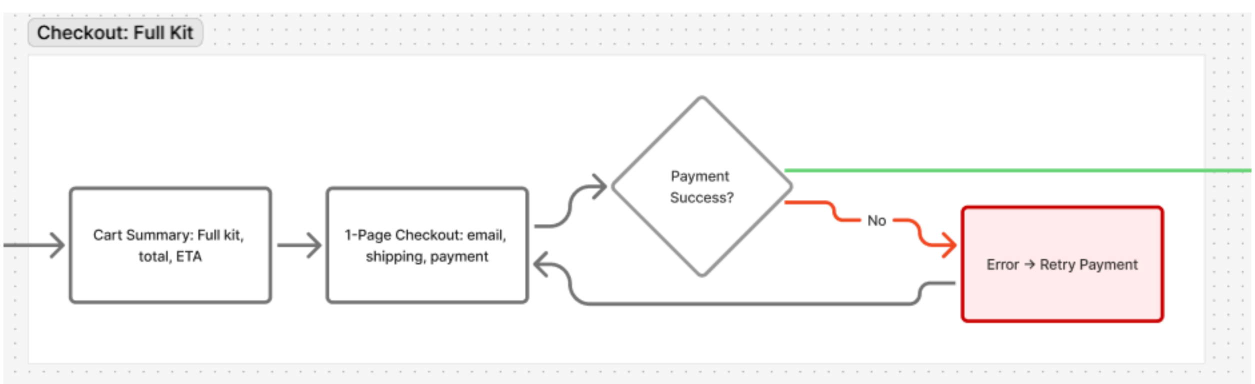
The following diagram illustrates all primary paths, decision points, and navigation patterns:

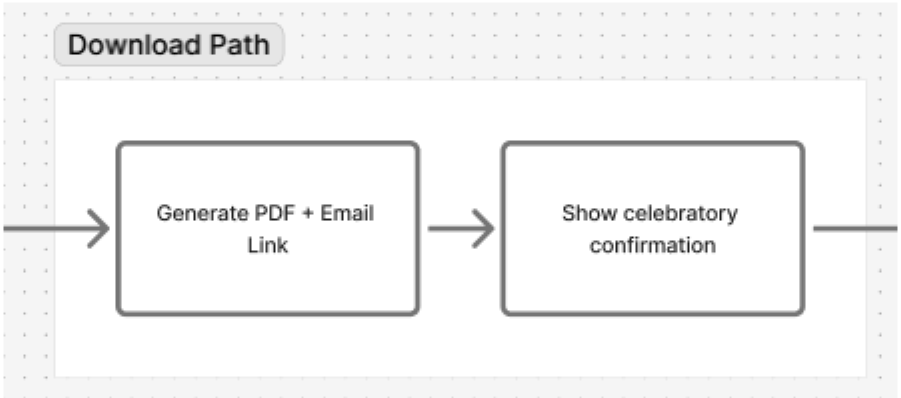


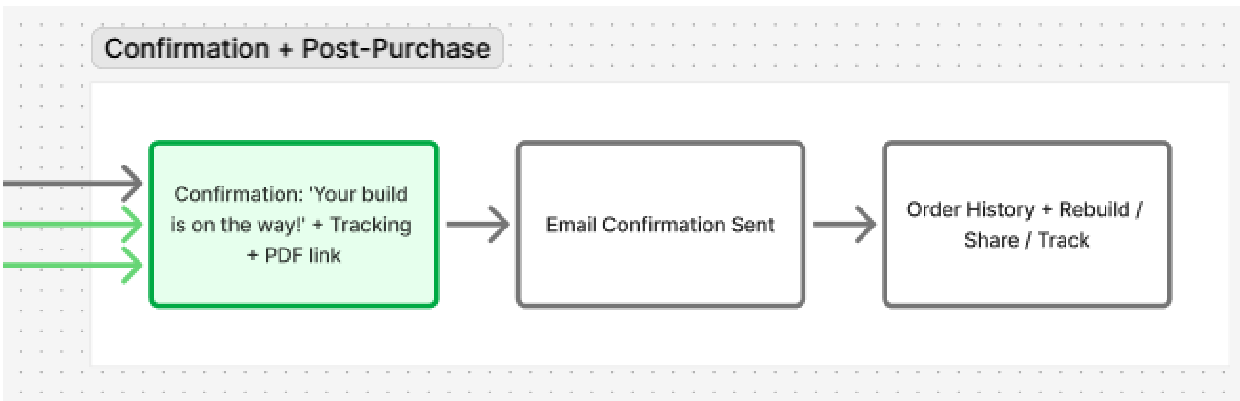

Edit & Refine Mode
Refine/Edit
Text box appears: 'Describe your changes...'
Suggestion Chips: Add detail, Simplify, Change color
Submit Edit → Spinner shows regeneration
Regenerat ion Succe...
No
Error: Friendly retry or back to preview
Yes


BOM & Pricing
Continue
Generate Bill of Materials
Show part images, names, colors, quantities
Show totals, price, availability, shipping
User Decision


Checkout: Missing Parts
Cart Summary: Missing parts, total, ETA
1-Page Checkout: email, shipping, payment
Payment Success?
No
Error → Retry Payment


Checkout: Full Kit
Cart Summary: Full kit, total, ETA
1-Page Checkout: email, shipping, payment
Payment Success?
No
Error → Retry Payment





Alternative Paths & Edge Cases

Upload Image Instead of Text:

- User clicks "Upload Image" tab on landing
- Drags/drops or selects image file
- System extracts concept and proceeds to generation

Unsatisfied with Model:

- User clicks "Try Different Prompt" from model viewer
- Returns to prompt input with previous text pre-filled
- Can modify and regenerate

Can't Afford Full Kit:

- User sees total price, feels hesitant
- System highlights "Free Download" option
- User gets PDF instructions, can source parts independently

Wants to Share Creation:

- User clicks "Share" icon (future phase)
- Gets shareable link or image
- Can post to social media

4. Screen-by-Screen Specifications

Screen 1: Landing / Prompt Input

Purpose: Capture imagination and guide users to input their first prompt

Layout Structure:

- Full-screen hero with example model gallery
- Prominent centered prompt input field
- Minimal navigation (logo, about link)
- Footer with trust signals

Element	Description	User Action	Emotional Note
Hero Headline	"Build Anything You Can Imagine"	Read	Inspiration
Subheadline	"Describe it. We'll generate the LEGO instructions."	Read	Clarity
Prompt Input	Large text field with placeholder: "A futuristic spaceship..."	Type/paste prompt	Anticipation
Generate Button	Prominent gradient button: "Generate My Build"	Click to start	Excitement
Example Gallery	4-6 rotating example builds with prompts shown	Browse, click for inspiration	Possibility
Image Upload Tab	Alternative input method	Upload image	Flexibility

Visual Hierarchy:

1. Hero headline (largest)
2. Prompt input field (most prominent)
3. Generate button (high contrast, impossible to miss)
4. Example gallery (supporting context)

Copy Examples:

- Placeholder text: "A medieval castle with dragons..." or "A tiny robot helper..."
- Button: "Generate My Build →"
- Helper text: "Be as specific or creative as you like"

Interactions:

- Input field expands slightly on focus
- Generate button pulses with subtle animation
- Examples cycle automatically every 5 seconds
- Clicking example pre-fills that prompt

Technical Notes:

- Input validation: minimum 3 words, maximum 200 characters
- Generate button disabled until valid input
- Smooth transition to loading screen

Screen 2: Generation / Loading

Purpose: Maintain engagement and build anticipation during AI processing

Layout Structure:

- Full-screen centered animation
- Progress indicator
- Dynamic status messages
- Optional: subtle background pattern

Element	Description	User Action	Emotional Note
Loading Animation	Animated bricks assembling or spinning blueprint	Watch	Wonder
Progress Bar	Linear bar filling 0-100%	Monitor	Anticipation
Status Messages	"Analyzing your idea...", "Selecting 247 bricks...", "Assembling model..."	Read	Engagement
Your Prompt	User's original prompt displayed at top	Remind	Ownership

Animation Sequence:

1. (0-3 sec) "Understanding your prompt..."

2. (3-6 sec) "Selecting optimal brick types..."
3. (6-9 sec) "Generating 3D model..."
4. (9-10 sec) "Finalizing instructions..."
5. (10 sec) Success animation + transition

Visual Treatment:

- Mechanical, satisfying animation style
- Bright, colorful brick elements
- Optional sound effects (clicking bricks)
- Smooth fade transition to model viewer

Copy Examples:

- Messages should feel specific and real
- Avoid generic "Please wait..."
- Use numbers when possible: "Analyzing 2,847 possible brick combinations..."

Interactions:

- No user interaction required (lean-back moment)
- Optional cancel button appears after 5 seconds
- Smooth crossfade to next screen

Screen 3: 3D Model Viewer

Purpose: Showcase the generated model and enable exploration

Layout Structure:

- 3D viewport occupies 70% of screen
- Right sidebar with model info and controls
- Bottom action bar with primary CTA
- Top navigation (logo, back button)

Element	Description	User Action	Emotional Note
3D Viewport	Interactive model with lighting, shadows	Rotate, zoom, pan	Delight
Model Name	AI-generated name: "Cosmic Speedster X-7"	Read	Personality
Stats Panel	Piece count, dimensions, difficulty	Read	Information

Rotation Controls	Visual affordances for interaction	Click/drag	Control
"View Instructions" Button	Primary CTA	Click to proceed	Readiness
"Edit Model" Button	Secondary action	Click to customize	Empowerment
"Try New Prompt" Link	Tertiary option	Start over	Safety net

3D Viewer Controls:

- **Rotate:** Click-drag horizontally/vertically
- **Zoom:** Scroll wheel or pinch
- **Pan:** Right-click drag or two-finger drag
- **Reset:** Double-click to return to default view

Model Presentation:

- Clean background (gradient or solid light color)
- Proper lighting to show depth and detail
- Subtle ambient shadows
- High-quality brick rendering (not blocky/pixelated)

Interaction Patterns:

- Model starts with gentle automatic rotation
- User interaction stops auto-rotation
- Helpful tooltip on first load: "Drag to rotate"
- Optional exploded view toggle

Copy Examples:

- Stats: "127 pieces | 15cm tall | Medium difficulty"
- Button: "View Building Instructions →"
- Helper text: "Drag to explore every angle"

Screen 4: Edit & Refine Controls

Purpose: Provide simple customization without overwhelming users

Layout Structure:

- 3D viewport remains visible (60% width)
- Left sidebar with edit panel (40% width)

- Preview updates in real-time or on click
- Clear save/cancel actions

Element	Description	User Action	Emotional Note
Color Palette	6-8 primary LEGO colors	Click to change model color	Creativity
Scale Slider	"Smaller ← → Larger"	Drag to adjust size	Control
Detail Level	"Simple / Balanced / Complex"	Select preference	Personalization
Regenerate Button	Apply changes	Click to update	Anticipation
Revert Button	Undo changes	Click to restore	Safety
Model Preview	Updated 3D view	Observe changes	Feedback

Edit Panel Sections:

1. Color Scheme

- Visual color swatches (red, blue, green, yellow, gray, black, white, custom)
- Clicking applies immediately to preview
- Optional: multi-color builds (advanced)

2. Size Adjustment

- Simple slider: -2 to +2 scale
- Shows estimated new piece count
- "Current: 127 pieces → Larger: ~230 pieces"

3. Complexity Level

- Three buttons: Simple | Balanced | Detailed
- Affects part count and build difficulty
- Visual examples of each level

Interaction Flow:

1. User makes adjustments via controls
2. Preview updates in real-time (if possible) or shows loading spinner
3. User clicks "Apply Changes"
4. Brief regeneration animation (3-5 seconds)

5. Updated model appears
6. User can continue editing or proceed

Visual Treatment:

- Edit panel slides in smoothly
- Active selections are highlighted
- Changes feel responsive and immediate
- Exit animation returns to full viewer

Screen 5: BOM & Pricing

Purpose: Present parts list and costs transparently, enabling informed purchase decisions

Layout Structure:

- Header with model summary
- Scrollable parts list (main content area)
- Right sidebar with pricing summary
- Bottom action bar with purchase options

Element	Description	User Action	Emotional Note
Model Preview	Small 3D thumbnail	Click to return to viewer	Context
Parts List	Table with images, names, colors, quantities	Scroll, review	Information
Individual Part Cards	Photo, LEGO ID, name, qty, price	Inspect	Transparency
Pricing Summary	Subtotal, shipping, total	Review	Anticipation
Availability Status	In stock / 2-day delivery	Read	Assurance
"Purchase Kit" Button	Primary action	Click to checkout	Commitment
"Download Free PDF" Link	Alternative action	Click to download	Freedom
"Modify Model" Link	Return to editor	Adjust before buying	Control

Parts List Table Format:

Thumbnail	Part Name	Color	Quantity	Unit Price	Subtotal
[Image]	Brick 2x4	Red	24	$0.15	$3.60
[Image]	Plate 1x2	Blue	48	$0.08	$3.84
...	...	...	...	...	...

Pricing Summary Box:

Parts Subtotal: $47.80
Shipping (USPS 2-day): $5.95

Total: $53.75

✓ All parts in stock
✓ Ships within 24 hours
✓ Free returns

Visual Treatment:

- Parts list uses card-based layout for scannability
- Part thumbnails are high-quality photos
- Colors are accurate to real LEGO pieces
- Pricing is prominently displayed but not alarming
- Trust signals reduce purchase anxiety

Copy Examples:

- Header: "Everything You Need to Build [Model Name]"
- CTA: "Purchase Complete Kit — $53.75"
- Alternative: "Download Instructions (Free)"
- Helper: "Price includes all 127 pieces plus building guide"

Interactions:

- Clicking part row expands detail view
- Hover shows larger part image
- Pricing updates if quantities change (future)
- Smooth scroll through parts list

Screen 6: Checkout & Confirmation

Purpose: Complete transaction smoothly and celebrate the purchase

Checkout Screen Layout:

Element	Description	User Action	Emotional Note
Order Summary	Model image, name, piece count, total	Review	Confirmation
Email Field	For order updates and instructions	Enter email	Connection
Shipping Address	Standard address form	Enter details	Practicality
Payment Method	Credit card or PayPal	Enter payment	Trust
Order Notes	Optional customization	Add notes	Personalization
Place Order Button	Final CTA	Click to complete	Commitment

Checkout Form Fields:

- Email address (required)
- Full name (required)
- Street address (required)
- City, State, ZIP (required)
- Phone (optional)
- Card number, expiry, CVV (or PayPal button)
- "Same as shipping" checkbox for billing

Visual Treatment:

- Single-page checkout (no multi-step)
- Progress indicator if multi-step (Email → Shipping → Payment)
- Sticky order summary on right side
- Clear field labels and validation
- Security badges and trust signals

Confirmation Screen Layout:

Element	Description	User Action	Emotional Note
Success Animation	Checkmark or celebration graphic	Watch	Joy
Order Number	Unique reference: #BB-20251101-0847	Note/copy	Assurance

Confirmation Message	"Your BrickBuilder kit is on the way!"	Read	Satisfaction
Delivery Estimate	"Arrives by Nov 8th"	Read	Anticipation
Digital Deliverables	"Download Instructions" button	Click to download	Immediate value
What's Next	Steps: tracking email, delivery, building	Read	Guidance
Social Share	"Share your creation" buttons	Click to share	Pride
Build Another	Return to landing	Start new project	Engagement

Confirmation Message Copy:

Your BrickBuilder Kit is Confirmed!

Order #BB-20251101-0847

We're preparing your 127 LEGO pieces for shipment.
Your custom building instructions are ready to download below.

Delivery: November 8, 2025
Tracking: Check your email for updates
Instructions: [Download PDF Now]

Start planning your build — instructions include tips,
estimated build time (2-3 hours), and a photo checklist!

Visual Treatment:

- Celebratory color scheme (confetti animation optional)
- Clear visual hierarchy prioritizes order number
- Download button is prominent and repeatable
- Friendly, encouraging tone throughout
- Next steps feel obvious

5. Visual Design System

Brand Identity

BrickBuilder Visual DNA:
Modern minimalism meets playful creativity. The design language should feel simultaneously sophisticated and approachable a tool for adults that remembers the joy of childhood creation.

We have not developed a color palette or a brand guideline in an effort to not stymie the creativity of the designer. Have at it and let's see what you can come up with.

6. Copy & Micro-Interaction Tone Guide

Brand Voice

BrickBuilder's voice is:

- **Encouraging:** "You're creating something amazing"
- **Clear:** "127 pieces, $53.75, ships in 2 days"
- **Playful:** "Let's build something awesome together"
- **Confident:** "We'll handle the technical stuff"

BrickBuilder's voice is NOT:

- Condescending or talking down
- Overly technical or jargon-heavy
- Pushy or sales-focused
- Cute to the point of annoying

Writing Principles

1. **Active Voice:** "Create your build" not "Your build can be created"
2. **User-Focused:** Use "you" and "your" liberally
3. **Specific Numbers:** "127 pieces" not "lots of pieces"
4. **Positive Framing:** "Choose your colors" not "You must select colors"
5. **Natural Brevity:** Short sentences. No fluff. Every word earns its place.

Headline Examples by Screen

Landing Page:

- Primary: "Build Anything You Can Imagine"
- Secondary: "AI-powered LEGO instructions in seconds"
- Alternative: "From Idea to Instructions to Reality"

Loading Screen:

- "Creating your masterpiece..."
- "Selecting the perfect bricks..."

- "Building something special..."

Model Viewer:

- Primary: "Meet Your Creation"
- Secondary: "[Model Name] — 127 pieces"
- Alternative: "It's Ready to Build"

Edit Screen:

- Primary: "Make It Yours"
- Secondary: "Customize colors, size, and detail"
- Alternative: "Perfect Your Design"

BOM Screen:

- Primary: "Everything You Need"
- Secondary: "127 pieces, ready to ship"
- Alternative: "Your Complete Building Kit"

Checkout:

- Primary: "Almost There"
- Secondary: "Your bricks ship within 24 hours"

Confirmation:

- Primary: "Your Build Is On The Way!"
- Secondary: "Instructions ready. Bricks shipping."

Button Copy Examples

Primary Actions:

- "Generate My Build"
- "View Building Instructions"
- "Purchase Complete Kit — $53.75"
- "Place Order"
- "Download Instructions"

Secondary Actions:

- "Edit Model"
- "Try Different Prompt"
- "Customize Colors"
- "Back to Model"

Tertiary Actions:

- "Learn More"
- "See Examples"
- "Share Creation"

Avoid:

- "Submit"
- "Continue"
- "Next"
- "OK"

Microcopy Examples

Input Placeholders:

- "A futuristic spaceship with blue engines..."
- "A medieval castle with towers..."
- "A tiny robot helper..."
- "Describe anything you want to build"

Helper Text:

- "Be as detailed or creative as you like"
- "Drag to rotate, scroll to zoom"
- "All parts guaranteed authentic LEGO"
- "Free shipping on orders over $75"

Loading Messages:

- "Understanding your vision..."
- "Selecting 247 optimal bricks..."
- "Calculating build instructions..."
- "Finalizing your model..."
- "Almost ready..."

Empty States:

- "No prompt yet — describe what you want to build"
- "Start by typing your idea above"

Error Messages:

- "Oops! Please describe your idea in a few more words"
- "We couldn't process that image. Try a different photo?"

- "Something went wrong. Want to try again?"

Tooltips & Hints

First-Time User Guidance:

- "New here? Type any idea and watch it become buildable LEGO instructions"
- "Drag to rotate your model"
- "Click any part in the list to see details"
- "All pieces are real LEGO parts you can actually buy"

Contextual Help:

- "Not quite right? Edit colors and size before purchasing"
- "Download instructions for free, or buy the complete kit"
- "Changing size affects piece count and price"

Emotional Micro-Moments

Delightful Copy That Appears at Key Moments:

After Generation Completes:

- " Boom! Your [model name] is ready"
- " Created in 8 seconds"

When User Rotates Model First Time:

- "Nice! Explore every angle"

On Adding to Cart:

- "127 pieces ready to ship "

After Purchase:

- " Amazing choice!"
- "Your instructions are downloading..."
- "Check your email for tracking"

If User Returns After Leaving:

- "Welcome back! Ready to create something new?"

Tone Adjustments by Context

Exploration Phase (Landing → Model View):

- Enthusiastic, inspirational
- "Imagine it. Build it. Make it real."

Decision Phase (BOM → Checkout):

- Clear, reassuring, transparent
- "All 127 pieces, $53.75 total, ships in 2 days"

Completion Phase (Confirmation):

- Celebratory, satisfied, forward-looking
- "Your kit is on the way! Instructions ready below."

Copy Don'ts

Avoid These Patterns:

- Technical jargon: "Initiating STL generation pipeline"
- Corporate speak: "Leveraging AI to facilitate..."
- Unnecessary apologies: "Sorry for the wait..."
- Vague promises: "Coming soon!" without specifics
- ALL CAPS (except rare emphasis)
- Excessive exclamation marks!!!!

Call-to-Action Best Practices

Structure: [Action Verb] + [Specific Object] + [Optional Value Prop]

Examples:

- "Generate My Build" (clear action + ownership)
- "Purchase Complete Kit — $53.75" (action + transparency)
- "Download Free Instructions" (action + value + cost)

Length:

- Primary buttons: 2-4 words
- Maximum: 6 words before reconsidering

Specificity:

- "Generate Model" → "Generate My Build"
- "Buy Now" → "Purchase Complete Kit"
- "Download" → "Download Instructions (PDF)"

7. Deliverables & Timeline

Designer Outputs

The Phase-1 MVP requires the following deliverables from the design team:

Deliverable 1: High-Fidelity Figma Prototype

Components:

- All 6 primary screens (Landing, Loading, Model Viewer, Edit, BOM, Checkout/Confirmation)
- Fully linked clickable prototype
- Desktop layout (1440px width optimized)
- 2-3 sample prompts with unique model outputs
- Interactive 3D model viewer mockup (simulated interactivity)
- Complete component library
- Design system documentation

Format: Figma file with organized pages and developer handoff specs

Acceptance Criteria:

- Prototype demonstrates complete user journey start-to-finish
- All interactions trigger appropriate transitions
- Design adheres to specified color palette, typography, spacing
- Components are properly documented and reusable
- User can complete flow in 2-3 minutes
- Prototype passes 5-second clarity test on each screen

Deliverable 2: Design System Documentation

Components:

- Color palette with hex codes and usage guidelines
- Typography scale with font files
- Component library (buttons, inputs, cards, etc.)
- Icon set with naming conventions
- Spacing and grid system
- Animation and motion specifications
- Accessibility annotations

Format: Figma design system + exported PDF reference guide

Acceptance Criteria:

- Any developer can implement components accurately from docs

- All colors, fonts, and spacing use systematic values
- Components include all states (default, hover, active, disabled, error)

Deliverable 3: User Flow Diagrams

Components:

- Complete end-to-end flow diagram
- Decision trees for alternative paths
- Edge case handling (errors, empty states)
- Navigation map

Format: Figma or Miro diagram + exported PNG/PDF

Acceptance Criteria:

- All possible user paths documented
- Decision points clearly marked
- Edge cases addressed

Deliverable 4: Screen Specifications Document

Components:

- Annotated screenshots of every screen
- Dimension specifications
- Content requirements
- Interaction notes
- Responsive behavior notes (if mobile included)

Format: Figma with annotations + exported PDF

Deliverable 5: Asset Package

Components:

- Example 3D model renders (3-5 models)
- Sample LEGO part images (20-30 parts)
- Icons (SVG format)
- Logo files (SVG, PNG)
- Any illustrations or graphics

Format: Organized ZIP file with clear naming conventions

Optional Deliverables (Stretch Goals)

Mobile Responsive Layouts:

- Tablet (768px) and mobile (375px) breakpoints
- Touch-optimized interactions
- Simplified mobile flows where necessary

Marketing Assets:

- Social media preview cards
- Email templates (order confirmation, shipping)
- Promotional graphics

Animation Prototypes:

- Loading animation video
- Transition demonstrations
- Micro-interaction examples

Research & Competitive Analysis

Recommended Research Activities:

1. **Competitive Audit**
    - LEGO Builder app
    - BrickLink Studio
    - Tinkercad
    - Other AI generation tools (DALL-E, Midjourney)
    - E-commerce checkout flows (Amazon, Shopify)
2. **User Research**
    - Interview LEGO enthusiasts about their building process
    - Survey potential users about willingness to pay
    - Usability test the prototype with 5-8 participants
    - A/B test key copy and layouts
3. **Technical Feasibility**
    - Validate AI model generation timeline (< 10 sec)
    - Confirm BrickLink API capabilities
    - Test 3D viewer performance across browsers
    - Assess part availability and pricing volatility

Brand Guidelines Extension

Voice & Messaging:

- Tagline: "Build Anything You Can Imagine"

- Mission statement: "BrickBuilder makes imagination tangible by transforming creative ideas into buildable LEGO creations in seconds."
- Brand personality: Innovative, Playful, Accessible, Magical

Logo Usage:

- Wordmark preferred over icon
- Clear space: minimum 24px around logo
- Minimum size: 120px width
- Color variations: Full color, white, black

Marketing Tone:

- Social media: Enthusiastic, community-focused, visual
- Email: Personal, helpful, transactional when needed
- Advertising: Aspirational, possibility-focused

Legal & Compliance Notes

Important Considerations for Development:

1. **LEGO Trademark:**
    - BrickBuilder must clarify it is not affiliated with LEGO Group
    - Use proper trademark symbols: LEGO®
    - Disclaimer: "LEGO® is a trademark of the LEGO Group, which does not sponsor, authorize, or endorse this product."
2. **Part Sourcing:**
    - Clearly communicate parts come from third-party marketplaces
    - No guarantee of official LEGO manufacturing
    - Quality disclaimers where appropriate
3. **User Content:**
    - Terms of service for uploaded images
    - Copyright considerations for generated models
    - Community guidelines for shared builds
4. **Age Restrictions:**
    - Compliance with COPPA (Children's Online Privacy Protection Act)
    - Age gates where necessary
    - Parental consent for users under 13
5. **Payment Processing:**
    - PCI DSS compliance

    - Clear refund and return policies
    - Data privacy (GDPR, CCPA)

Metrics & KPIs (Future)

When MVP transitions to functional product, track:

Engagement Metrics:

- Prompt submission rate
- Model generation completion rate
- Time spent in 3D viewer
- Edit feature usage rate

Conversion Metrics:

- Prompt → Model view: ___%
- Model view → BOM view: ___%
- BOM view → Checkout: ___%
- Checkout → Purchase: ___%

Business Metrics:

- Average order value (AOV)
- Customer acquisition cost (CAC)
- Lifetime value (LTV)
- Repeat purchase rate

User Satisfaction:

- Net Promoter Score (NPS)
- Customer satisfaction (CSAT)
- Task completion rate
- Support ticket volume

Glossary of Terms

AI Generation: The automated process of creating 3D LEGO models from text or image prompts using machine learning

BOM (Bill of Materials): Complete list of LEGO parts required to build the generated model, including quantities and part IDs

Build Instructions: Step-by-step visual guide showing how to assemble the LEGO model

LDR Format: LDraw file format, standard for digital LEGO designs

Part ID: Unique identifier for specific LEGO pieces (e.g., "3001" for a 2×4 brick)

Prompt: User input (text or image) that describes the desired LEGO creation

STL File: 3D model file format, useful for 3D printing or advanced editing

Studs: The round bumps on top of LEGO bricks that enable connection

Contact & Resources

For Questions or Clarifications:

- Product Lead: [Name/Email]
- Design Lead: [Name/Email]
- Technical Lead: [Name/Email]

Useful References:

- LEGO Design Guidelines: [URL]
- BrickLink API Documentation: [URL]
- Figma Design File: [URL]
- Project Management Board: [URL]

Internal Resources:

- Brand Assets Folder: [Drive Link]
- Research Findings: [Document Link]
- Stakeholder Presentation: [Slide Deck Link]

Conclusion

BrickBuilder represents an ambitious vision: to create the world's first seamless imagination-to-reality platform for LEGO building. This Phase-1 MVP PRD provides the foundation for a design that is bold, clear, and user-centered.

The success of this prototype hinges on three critical elements:

1. **Clarity of Purpose:** Every screen must communicate value instantly
2. **Emotional Resonance:** The experience must feel magical, not mechanical
3. **Attention to Detail:** Polish and craft signal quality and care

When executed well, this prototype will not only validate the product concept but also inspire confidence in stakeholders, excite potential users, and provide a clear roadmap for engineering implementation.

The goal is simple: Make someone say "I need this" within 30 seconds of seeing it.

Let's build something amazing.

BrickBuilder Phase-1 MVP Product Requirements Document
Version 1.0 | October 24, 2025 - Rev1
Prepared for Design Team Review